IB Mathematics AA SL — Calculus Review Worksheet

Total: 20 marks. Give non-exact answers to 3 significant figures.

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1 Given $f(x) = 3x^4 - 5x^2 + 2x$, find $f'(x)$. **[3]**

2 Solve $2^{x+1} = 48$. **[3]**

3 A geometric series has first term 6 and common ratio $\frac{1}{3}$. Find its sum to infinity. **[2]**

4 Evaluate $\displaystyle\int_0^2 (3x^2 + 1)\,dx$. **[3]**

5 Find $\displaystyle\int xe^x\,dx$ using integration by parts. **[4]**

6 Find the x-coordinates of the stationary points of $y = x^3 - 6x^2 + 9x + 1$. **[4]**

Answer key

1. $f'(x) = 12x^3 - 10x + 2$
2. $x = 4.59$
3. 9
4. 12
5. $xe^x - e^x + C$
6. $x = 1$, $x = 3$